Round-3 review: *The worker is the constant-branch slice of the weld*

(MacBeth, email UID 418, content commit `a452538`; follows my round-2 report on UID 412)

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Date:	2026-10-19
To:	Scot MacBeth (Kodamai); cc Robin
Reviewed:	`scotmacbeth/work-in-progress`, `papers/2026-10-18-weld-worker-bicomodule.pdf` @ `a452538` (10 pp.; attachment byte-identical to the repo copy at HEAD)
This review:	Rick WIP commit `85ecf0d`
Check script:	`scripts/day247/macbeth418/check_weld_r3.py` (+`.log`), reuses the round-2 engine, <2 s

Notation as yours: $n = |S|$, $m = |T|$, $B = \mathsf{St}_S \triangleleft \mathsf{St}_T$, $B' = S \times \mathrm{const}(T) \cong A$.

Verdict

- **Round-2 M1 (retract): LANDED in the text.** Theorem 6 is now explicitly position-set level. Prop. 2 states that e fails the λ-square, Theorem 7 installs r_{s_0}, and Theorem 8 and §8 agree. Recomputed: e fails the λ-square on *exactly* the non-constant positions (180 of 192 at $(3,4)$, 312 of 324 at $(4,3)$, 40 of 50 at $(2,5)$) and passes the ρ-square everywhere. r_{s_0} is a bicomodule retraction onto B' at *every* base point $s_0 \in S$, with fibre m^{n-1}, at all six shapes below.
- **Round-2 M2 (clause-2 readings): landed in the Verdict paragraph, Thm 8 and §8, but NOT throughout.** Two sentences elsewhere still assert the conflated reading and contradict your own verdict. See new M1.
- **Demoted clauses at generic shapes.** I tested $(n,m) \in \{(1,2),(1,4),(4,1),(3,4),(4,3),(2,5)\}$ and computed $\bar{F}_A = mV^n$, $\bar{F}_B = m^nV^n$, worker = content iff $m = 1$, and worker = weld iff carriers equal iff $n = 1 \vee m = 1$. All agree with your Verdict paragraph. The separating shapes are $(1,m)$ with $m \geq 2$, where "content" and "weld" part ways. The generic shapes $n, m \geq 2$ only confirm the counts, because both readings fail there. A test at those shapes alone would not separate the readings.
- **Grades.** Thm 5, the clause-2 Verdict paragraph, Thm 6, Prop. 2 and Thm 7 are **checked-sober** (by hand, and computed at the shapes above). The general $\pi_1 A \cong B'$ identity (Remark 3) is computed at all six shapes, both squares. You have it as "not written out", but it is proved by hand in my round-2 S2, so it is three lines away from proved (S1). Registry: clause 3 → proved is fine. Clause 2 → proved is fine *for Thm 5 and the Verdict paragraph*, once M1 is fixed.
- **New: MUST 1, SHOULD 1, NIT 2.**

MUST

M1. The M2 conflation survives in Remark 4 and the Introduction.

- Remark 4, end (p. 7, tex l. 510–511) reads: "at $|S| = 1 \ldots$ There the worker *is* the content, on the nose, and no negative verdict may be claimed." Your Verdict paragraph, half a page later, says that worker = content factor iff $|T| = 1$, and that at $(1,2)$ $\bar{F}_A = 2V \neq V$. Recomputed: at $(1,2)$, $\bar{F}_A(3) = 6 \neq 3$, and at $(1,4)$, $\bar{F}_A(3) = 12 \neq 3$. At $|S| = 1$ the worker is the whole weld, not the content, and the negative verdict on "worker = content" *is* claimable whenever $m \geq 2$.
- Introduction (p. 2, tex l. 136) reads: "the naive comparison gives the wrong locus, declaring worker ≠ content already at $|S| = 1$, where in fact $B = A$". Under the content-factor reading that declaration was *right* at $(1, m \geq 2)$. What was wrong was the inference from $B = A$.
- Example 1, last sentence (tex l. 535–536) contrasts "$|T| \geq 2$" with "the corrected criterion $|S| \geq 2 \wedge |T| \geq 2$" without saying which equality each criterion is for. By your own Verdict paragraph, $|T| \geq 2$ is the correct criterion for worker ≠ content, and $|S|, |T| \geq 2$ is the criterion for worker ≠ weld.

Fix. In Remark 4, replace the last sentence with "There the worker is the whole weld; it is the content factor only if also $|T| = 1$." Reword the Introduction sentence and label the two criteria in Example 1. These are three sentences in total, and no theorem changes.

SHOULD

S1. Close Remark 3 instead of deferring it. The proof of Prop. 1 says "the two structures agree (Remark 3)". Remark 3 then says the general identity is "not written out" and only checked at five shapes. A proof that cites a check is graded computed. The general identity is the explicit computation in my round-2 S2: $\lambda_A(s,t) = (s, a \mapsto (a,t))$, $\lambda_A^\sharp(a,(a',t')) = (a',t')$, $\rho_A(s,t) = ((s,t),(a,t') \mapsto t')$, $\rho_A^\sharp((a,t'),t'') = (a,t'')$. Applying σ gives the λ,ρ of B' with $g = \mathrm{const}_t$. Paste it in and delete Remark 3. Prop. 1 is then proved for all S,T. (Computed here at all six shapes: $\pi_!A$ satisfies all five laws, and σ commutes with both squares.)

NIT

N1. Thm 7 and the sentence after it say "e, r_{s_0} induce the same map on $\bar{F}$". Prop. 2 shows that e is not a bicomodule map, so it induces no map of prafunctors. Say instead that they agree on the fibre $\bar{F}(V)$ evaluated at the base object s_0, or drop the clause. My round-2 wording seeded this, so the slip is mine as much as yours.

N2. "The method" (p. 3, tex l. 178): "the coactions preserve constancy because they never touch the branch-map coordinate". This is true, but it is the mechanism you retired in Remark 2: every slice is preserved, and closure comes from S-saturation. Point to Lemma 3(b) instead.

Script output (abridged; full log in repo)

```
(1,2) F_A=6,  F_B=6,    content=3  worker=content False [pred |T|=1: False]; worker=weld True  [pred True]
(1,4) F_A=12, F_B=12,   content=3  worker=content False; worker=weld True ; Rmk4 claim "|S|=1 => content": False
(4,1) F_A=81, F_B=81,   content=81 worker=content True ; worker=weld True
(3,4) F_A=108,F_B=1728, content=27 both False; e lam-square fails on exactly non-const positions (180)
(4,3) F_A=243,F_B=6561, content=81 both False; e fails on exactly non-const (312)
(2,5) F_A=45, F_B=225,  content=9  both False; e fails on exactly non-const (40)
all shapes: B laws, pi_!A laws, sigma both squares, e rho-square, r_s0 retraction (every s0), fibre m^(n-1): True
ALL ROUND-3 ASSERTIONS PASSED
```